

BÙI HỒNG PHÚC (Ethan)

📍 Ho Chi Minh City, Vietnam 📞 0379 517 012 ✉ duyphuc2425@gmail.com 🌐 github.com/Phucgiacat
in LinkedIn

Technical Skills

Programming: Python (Proficient), C/C++, SQL

AI/LLM: LangChain, RAG, Prompt Engineering, HuggingFace Transformers, OpenAI/Anthropic APIs

ML & Data: PyTorch, NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib

MLOps & Deployment: Git, Docker, Flask, REST APIs

Languages: English (Professional), Vietnamese (Native)

Experience

NLP Research Lab, University of Science — Research Assistant

Feb 2025 – Present

- Built an OCR-YOLOv11 auto-labeling pipeline using Levenshtein matching, reducing manual correction time by **60%**.
- Developed web-based annotation tools for OCR validation, improving team throughput.
- Researched and implemented LLM optimization techniques for Vietnamese machine translation tasks.
- Onboarded and mentored new lab members; maintained team codebase using Git workflows.

CMT Global — Technology Development Intern

Feb 2025 – Apr 2025

- Debugged and resolved production software issues to maintain system stability.
- Executed large-scale data crawling scripts for internal analytics.
- Performed functional and manual testing for product verification.

Projects

Document Visual Understanding with OCR-VLM (*CoverGO Hackathon 2025*)

Oct 2025 – Nov 2025

- Developed a multimodal OCR-Vision-Language model for Vietnamese receipts and forms.
- Built semi-automatic layout parsing with PaddleOCR and Gradio.
- Fine-tuned Donut and Qwen-VL for key-value extraction tasks.
- **Tech:** PyTorch, Transformers, Donut, Qwen-VL, Gradio
- **Project:** github.com/Phucgiacat/CoverHackatho-project 🔗

Hallucination Detection (*Data Science Challenge 2025*)

Aug 2025 – Oct 2025

- Designed a multi-head reward model to detect factual hallucinations in multimodal LLM outputs.
- Implemented evaluation pipelines to measure hallucination rates automatically.
- Applied GRPO on LoRA-adapted models, achieving **86% accuracy** on the test set.
- **Project:** github.com/monster1909/Detect-hallucination-using-LoRa-and-GRPO 🔗

Automatic Data Labeling Pipeline (*Personal Project*)

Feb 2024 – Aug 2024

- Built an end-to-end automatic labeling system integrating OCR and YOLOv11.
- Achieved **90%+ character-level accuracy** and reduced annotation workload by over **60%**.
- Published paper accepted at **VCL 2025**.
- **Project:** github.com/Phucgiacat/ocr_corrector 🔗

Certifications & Achievements

- 2nd Prize – Bosch Innovation Day 2025
- Coursera: Supervised ML & Advanced Learning Algorithms – DeepLearning.AI (2024)
- Python (Basic) – HackerRank (2024)
- 3rd Prize – Provincial Science Fair
- Participant – UIT Data Science Challenge 2025
- **Publication:** Paper accepted at VCL 2025 (Vietnamese Conference on Linguistics)
- [Online Portfolio & Certificates Available](#) 

Education

University of Science, Ho Chi Minh City

Sep 2022 – Present
(Expected 2026)

- Bachelor of Information Technology
- **GPA: 7.98/10**
- **VSTEP: B1**